

Backup slides

Md. Sakif Khan 0421052099

- 2 Budget configuration and enforcement sites
- 3 Which budgets actually bind
- 4 The full failure census
- 5 Hedging behaviour, all five systems

Backup: budget configuration

Parameter	Value	Enforced in
max_depth	4	both routers
beam_width	3	explorer (slice)
max_anchors	5	explorer <i>and</i> evaluator
max_backtracks	3	post-evaluation router
max_llm_calls	25	LLM client
max_verify_iters	2	post-verification router, verifier
max_seconds	300	both routers

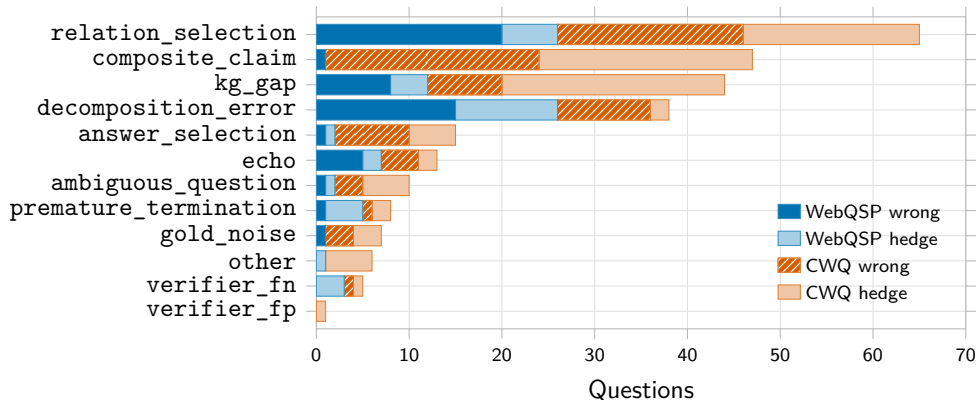
Termination does not depend on model behaviour: every cycle passes through a router that checks a monotone counter.

Backup: which budgets actually bind

Budget	WebQSP	CWQ	Pooled
Depth cap	16.5%	34.8%	25.6%
Backtrack cap	3.2%	9.2%	6.2%
Verify-iteration cap	1.0%	2.8%	1.9%
Call cap	0.0%	0.0%	0.0%

Percentage of questions on which each budget refused a request. AGR never reaches the 25-call cap — which is what makes the comparison against a clipped Think-on-Graph meaningful.

Backup: the full failure census

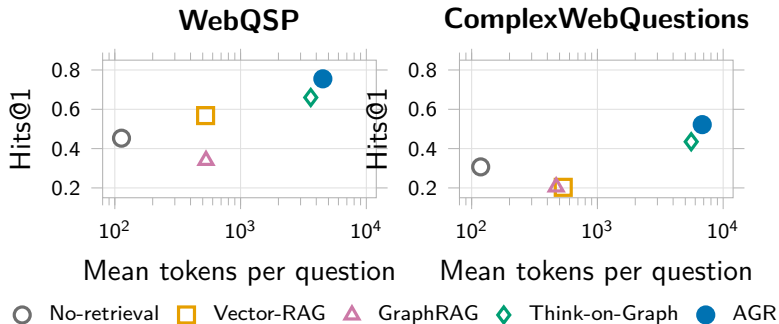


Backup: hedging behaviour

System	WebQSP hedge %	CWQ hedge %
No-retrieval	12.2	38.0
Vector RAG	27.0	48.2
Static GraphRAG	55.8	60.8
Think-on-Graph	18.8	22.5
AGR	8.2	23.0

A hedge is a refusal to assert, not a wrong answer. AGR hedges least on WebQSP while also being most accurate there.

Backup: accuracy against cost



On *tokens* Think-on-Graph is **not** dominated — it buys lower accuracy at a lower token price. On *calls*, which the budget meters, it is.

The frontier depends on which cost you charge.

Backup: the benchmark was wrong 57 times

The same cross-system agreement is also a **detector**: consensus flagged 105 questions, adjudication confirmed 41, and the gap is the attractor above rather than label noise.

- 41 excluded before the census
- 17 found inside the census
- 1 counted in both

⇒ 57 **distinct questions**

Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

A contribution that outlives this particular system.